The Economics of Orbit Use: Open Access, External Costs, and Runaway Debris Growth

Akhil Rao, Giacomo Rondina

Abstract: We present a dynamic physico-economic model of Earth orbit use with endogenous satellite collision risk to study conditions under which debris-producing collisions between orbiting bodies result in debris growth that may render Earth's orbits unusable, an outcome known as Kessler syndrome. We characterize the dynamics of objects in orbit under open access as well as when external costs—the impact of an additional satellite launch on the collision risk faced by all satellites—are internalized, and we show that Kessler syndrome can emerge in both cases. Finally, we show that once the economic incentives of satellite launching are modeled, for Kessler syndrome to emerge, autocatalytic debris growth is essential. In our main calibration, Kessler syndrome can emerge anytime between the year 2040 and the year 2184, with the precise date being very sensitive to the calibration of autocatalytic debris growth parameters.

JEL Codes: D62, Q20, Q54, Q57

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SATELLITE SERVICES ARE INCREASINGLY IMPORTANT in the modern world. As humans launch more satellites, the risk of collisions between orbiting objects

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increases. Such collisions can destroy satellites and produce orbital debris, further increasing the risk of future collisions. The worst-case scenario is runaway debris growth, known as Kessler syndrome, wherein the production of debris due to collisions between orbiting bodies becomes self-sustaining and irreversible. Kessler syndrome will render valuable regions of orbital space unusable for decades, centuries, or longer. To make matters worse, international treaty law places orbital space under an open-access regime: anyone can place a satellite in any orbit they choose.

How will open access affect orbital debris accumulation, collision risk, and the occurrence of Kessler syndrome? What are the external costs that create a coordination problem under open access? Can Kessler syndrome result even if external costs are fully internalized? To address these questions, we study a dynamic model that combines the economic intertemporal decision problem of satellite operators with the physics of orbiting objects.

We identify a key economic condition that determines the optimal satellite launch rate. The condition equates the cost of launching a satellite to its return, discounted by an "effective discount rate" that reflects time-varying collision risk. As satellites and debris jointly evolve over time, so does the collision risk and the resulting launch rate, establishing a feedback loop where the economic incentives shape the physical dynamics. Under open access satellite operators do not internalize the full impact of their launching decisions on the collision risk, leading to an inefficiently high launch rate. We show that the resulting external cost can be represented as an upward adjustment to the effective discount rate that a social planner would implement in choosing the size of an optimal satellite fleet.

Our analysis yields three key findings. First, under open-access conditions, Kessler syndrome is a significant issue only with autocatalytic (self-reinforcing) debris growth. Without this growth, dynamic payoff-maximizing launch decisions usually lead to sustainable orbits for satellites and debris. Second, we find that Kessler syndrome can still emerge even when external costs—the impacts of private launch decisions on collision risks and debris accumulation—are entirely accounted for. This phenomenon arises from the dominance of short-term gains of a larger satellite fleet over the long-term benefits of a smaller one, illustrating the intertemporal trade-off at the heart of our economic model. Third, our study indicates that the timing of Kessler syndrome under open access is highly variable, potentially occurring between 2040 and 2184. This broad range highlights the system's sensitivity to the dynamics of autocatalytic debris growth and changing satellite return rates, a central insight of our theoretical analysis.

Institutional Background. To lay the groundwork for our model, we describe key institutional and physical aspects of orbit use. The 1967 Outer Space Treaty (OST) is the main legal framework governing orbit use. It prohibits the establishment of orbital property rights, effectively making outer space an open-access commons.[1] This lack of

1. Specifically, Article 2 of the OST forbids national appropriation or claims of sovereignty over outer space, prohibiting national authorities from establishing orbital property rights (Gorove 1969).

property rights means that satellites often cluster around valuable orbital positions, risking collisions, similar to congestion issues in high-seas fisheries (Gordon 1954).[2]

Satellites also produce debris over their life cycle. Launches leave spent rocket stages and separation bolts in orbit, satellite operations leave paint chips and tools behind, and satellites that are not deorbited or shifted to disposal orbits at the end of their life become debris themselves. Satellites colliding with debris or other satellites can shatter into hundreds of hazardous debris fragments.[3] Compounding the problem, collisions between debris objects can generate even more hazardous debris. Debris accumulation can cause a cascading series of collisions between orbital objects, creating a growing debris field that can render orbital regions unusable and impassable for decades or centuries. This phenomenon is known as "Kessler syndrome" (Kessler and Cour-Palais 1978).[4]

Connection to the Literature. The interaction between the physical dynamics and the intertemporal economic incentives to launching are underexplored in the literature on orbit use.[5] We contribute to the literature on decentralized and optimal responses to

2. Ostrom et al. (1999) provide some insight into why decentralized orbit management may face challenges. Orbit users are a diverse and international group, ranging from national militaries and intelligence agencies to corporations, universities, and wealthy individuals. Physically excluding potential users from orbital regions without creating additional debris is difficult, and the relevant conflict resolution mechanisms are unclear. National regulatory regimes must also contend with "launch leakage," which has happened at least once already (Selk 2017). Weeden and Chow (2012) discuss some of these issues in more detail.

3. Objects in orbit move at velocities higher than 5 kilometers per second (km/s), so even debris as small as 10 cm in diameter can be hazardous to active satellites. Debris as low as 900 km above the Earth's surface can take centuries to deorbit naturally, while debris at 36,000 km can take even longer (Weeden 2010). Currently, there are more than 2,000 operating satellites in orbit, up to 600,000 pieces of debris large enough to cause satellite loss, and millions of smaller particles that can degrade satellite performance (Ailor et al. 2010).

4. Kessler syndrome can cause large economic losses, both directly from damage to active satellites and indirectly from limiting access to space (Bradley and Wein 2009; Schaub et al. 2015). Existing estimates of debris growth indicate that the risk of Kessler syndrome is highest in low Earth orbit (LEO), where it threatens current and future imaging and telecommunications satellites and can reduce access to higher orbits (Kessler et al. 2010; Lewis 2020). In the worst-case scenario Kessler syndrome could completely block human access to space, marking an eventual end to services like GPS and satellite imaging. Long-run disruption of satellite services will make it harder to measure economic activity, reduce weather-related uncertainty, measure environmental degradation and respond to natural disasters, monitor environmental policy compliance, and meet conservation goals (Donaldson and Storeygard 2016; Sullivan and Krupnick 2018; Baragwanath et al. 2019; O'Connor et al. 2019; Cooke and Golub 2020; Jain 2020; Stroming et al. 2020; Bernknopf et al. 2021).

5. Wienzierl (2018) highlights several issues in the development of a space economy, from space debris to holdup problems and market design, which economists are uniquely positioned to address. Our focus is solely on the environmental problems in orbit associated with existing institutions.

interacting dynamic externalities by showing how profit-maximizing responses to congestion of the commons can limit pollution production, and physico-economic conditions under which profit maximization leads to runaway pollutant accumulation.[6] Though prior economic literature on orbit use has identified and quantified the externalities driving collisional rent dissipation (Adilov et al. 2015; Béal et al. 2020; Rao et al. 2020; Rouillon 2020), the channels through which the externality operates have not been explored in detail. Further, the economics of Kessler syndrome remain understudied, with prior literature either explicitly neglecting it (e.g., Rao et al. 2020; Rouillon 2020) or considering simplified dynamics (e.g., Adilov et al. 2018) or different social planner notions (Bongers and Torres 2023).[7] Our framework unifies existing research on orbit use, showing model features necessary to obtain different outcomes and enabling calibration and quantitative analysis. We also identify an important difference between resource-use dynamics when the relevant capital stock is provided by nature (e.g., fish in fisheries) and when it is provided by humans (e.g., satellites in orbits): while increases in the discount rate make equilibrium and optimal resource collapse more likely for natural capital, they have the opposite effect for artificial capital.

The rest of the paper is organized as follows. In section 1, we present a simple model that develops the essential economic intuition for orbit use. In section 2, we present a physical model that captures the key elements of orbits as a resource and shows how Kessler syndrome can occur. Section 3 develops the dynamic economic model of open-access and optimal orbit use in the general physical setting, decomposing the channels through which the externality operates and illustrating how Kessler syndrome shapes the dynamics of economic orbit use. We connect our results to the literature on natural

6. While open-access problems have been well studied in terrestrial settings such as fisheries, forests, climate, oil fields, traffic, and invasive species management, open access to orbital resources is not as well understood. Though results from these other settings provide some helpful intuition, open access and the orbital mechanics governing collision risk and debris production create unique physico-economic feedback loops.

7. Adilov et al. (2018) consider a model environment in which satellites could become economically unprofitable before they become physically unusable (satellites are destroyed with probability one), a situation they describe as "economic Kessler syndrome." In so doing they do not explore the dynamics of physical Kessler syndrome. We define Kessler syndrome as states from which the debris stock diverges to infinity ("runaway debris growth") and allows collisions between debris objects. The latter feature is critical for understanding orbit-use dynamics, as collisions between debris are becoming increasingly likely and are expected to dominate the long-run dynamics of the orbital environment (Davenport 2020; Lewis 2020). Bongers and Torres (2023) compare orbit use with and without debris risk, finding that debris risk reduces the equilibrium number of satellites. They consider a notion of Kessler syndrome similar to ours but do not analyze dynamics of Kessler syndrome under open access and a social planner when both face collision risk.

resource exploitation and conclude in section 4. Proofs, derivations, and extensions are shown in the appendixes (apps. A–D are available online).

1. A SIMPLE MODEL OF ORBIT USE

We begin by presenting a stylized model of satellite launch and operation with the goal of isolating the interplay between the economic and physical mechanisms in orbital use. Time is finite and divided into three periods, $t = 0, s, l$. In period 0 launches take place; in period s and l the satellites operate and produce revenue. Between periods, satellites face the risk of collision, which crucially depends on the total number of satellites in orbit.

1.1. Orbit

Consider a spherical shell ("orbit") around the Earth divided into a finite number of slots, each slot able to hold one satellite. Once in orbit, due to unpredictable perturbations, satellites can collide with each other or with orbiting debris. Upon collision, the satellites become inoperable but remain in orbit in the form of debris.

Let $X \in \mathbb{R}$ denote the objects in orbit at a given period, whether operating satellites or debris.[8] The probability of a satellite surviving into the next period depends negatively on the total objects in orbit and is denoted by $q(X) \in [0, 1]$, with $q'(X) \leq 0$. We assume that there exists a $\bar{X} < \infty$ such that $q(X) = 0$ for all $X \geq \bar{X}$. Intuitively, $\bar{X}$ corresponds to the number of orbiting objects at which the orbit becomes unusable because any satellite would not survive after launch with probability 1.

We assume that launches happen only in period 0.[9] The stock of objects in orbit at the end of period 0 always corresponds to the satellite launched at time 0, which we denote by $S \in \mathbb{R}$. The probability of a satellite operating in period s corresponds to its probability of not colliding with any objects, so there are $q(S)S$ satellites operating in the short run. When a satellite collision occurs, either with another satellite or debris, fragments are produced so that more than one individually orbiting object results. We assume that fragments in excess of 1 form at the rate $\sigma > 0$.[10] The number of satellites that collide between periods 0 and s is $(1 - q(S))S$, and they generate $(1 + \sigma)(1 - q(S))S$ units of debris. The number of orbiting objects at the end of period s, which we denote by $g(S)$, is thus the sum of the satellites operating in period s and the debris formed from the collisions. Summing the two and rearranging one obtains

8. In our analysis, we treat each individual satellite launch as an infinitesimal increment to X. None of our results depend critically on this, while their mathematical characterization is substantially simplified.

9. This assumption is made for simplicity. In the full dynamic model, launches can happen in any period.

10. For simplicity, in this setting we do not explicitly model debris-debris collisions or natural decay of debris from orbit, though we incorporate both features separately in our full model.

$$g(S) \equiv S + \sigma(1 - q(S))S. \tag{1}$$

The formation of fragments from collision, that is $\sigma > 0$, implies that the total number of objects in orbit increases between the end of period 0 and the end of period s, that is $g(S) > S$. Assuming that collision probabilities are independently distributed, the probability that a satellite operates in period l, evaluated at the time of its launch, is thus $q(S)q(g(S))$.

1.2. Economic Returns

The launch of a satellite requires a unitary cost, $F > 0$, to be incurred at time 0. The satellite, conditional on being operational, generates a constant per-period return, $\pi > 0$, in periods $t = s, l$, discounted at the per-period rate r. The model can be generalized to allow for the return π to be a function of S, and we do so in appendix A for the simple model, and in the simulation exercise in section 3.4 for the full dynamic model. While the dependence of π on S has implications for the quantitative simulation, the qualitative features of our results do not critically depend on it.

Let $V(S)$ denote the expected present value of the lifetime net return from a satellite when S satellites are launched. This can be written as

$$V(S) = -F + \frac{\pi q(S)}{1 + r} + \frac{\pi q(S)q(g(S))}{(1 + r)^2}. \tag{2}$$

The structure of the discounted expected return at time 0 in equation (2) is intuitive: the expected return in period s is $\pi q(S)$, while in the long run it is $\pi q(S)q(g(S))$, both appropriately discounted. The marginal expected profits of an additional satellite are decreasing in the total number of satellites launched. Intuitively, an additional satellite launched increases the number of objects in orbit in both the short run and the long run, hence decreasing the survival probability in both periods. Additionally, equation (1) suggests that the long-run negative impact of an additional satellite on the survival probability is increasing in σ.

1.3. Equilibrium with Open Access

We consider first the equilibrium choice of satellite launches, S, under open access. We assume that property rights over slots cannot be assigned—satellite operators own the satellites they launch but not the slots they occupy. Lacking slot rights, operators cannot exclude each other from taking unoccupied slots despite the collision risks posed.

Under open access, launches take place until the expected profits are equal to zero, that is, $V(S) = 0$. Using equation (2) with equation (1), the open-access equilibrium launches, $\hat{S}$, correspond to the level at which the private cost of a launch is equal to the expected revenue from the satellite:

$$V(\hat{S}) = 0 \Rightarrow \underbrace{F}_{\substack{\text{Private cost}\\ \text{of a satellite}\\ \text{(PC)}}} = \underbrace{\pi \frac{q(\hat{S})}{1+r}\left(1 + \frac{q(g(\hat{S}))}{1+r}\right)}_{\substack{\text{Expected revenue}\\ \text{from a satellite}\\ \text{(ER)}}}. \tag{3}$$

Since $q(0) = 1$ and $g(0) = 0$, an open-access equilibrium with positive launches, that is, $\hat{S} > 0$, occurs whenever the cost of launching is smaller than the expected return of the first satellite launched, that is, $F \leq \pi[1/(1+r)](1 + [1/(1+r)])$. We assume that this condition holds. Note that in choosing whether to launch, an individual agent takes S as given and thus does not account for the impact that their additional launch will have on the probability of survival of the entire satellite fleet.

Figure 1 shows the diagram representation of the economic problem of a satellite operator. The PC curve represents the private cost of launching, which is constant in our model and equal to F. The ER curve represents the expected lifetime return of one satellite, which corresponds to the right-hand side of (3). The expected return is decreasing in the number of satellites in orbit, S, due to the lower survival probability when S is larger. The difference in the ER curves between panels A and B reflects the higher σ (1.25 vs. 20): for any level S, the increase in debris formation increases the

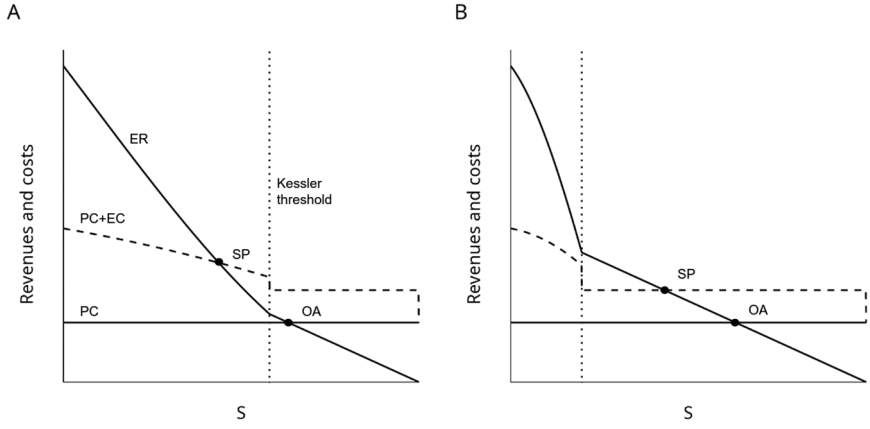


Figure 1. Equilibrium diagram for the simple model. Curves are labeled as in equations (3) and (4). Curve ER shows the expected revenue of one additional satellite as a function of S, PC shows the private cost of one additional satellite, and the dashed curve PC + EC shows the social cost of one additional satellite (i.e., the sum of private and external costs). The survival probability is specified as $q(X) = 1 - X/\bar{X}$. The parameters are set to $\pi = 1$, $r = 0.05$, $F = 0.35$, $\bar{X} = 5$, with $\sigma = 1.25$ in panel A and $\sigma = 20$ in panel B. The dotted vertical lines show the Kessler threshold. Under the chosen parameterization, open access causes Kessler syndrome in both panels A and B, while the social planner only causes Kessler syndrome in panel B.

objects in orbit and thus reduces the survival probability between the short and the long run, shifting the ER curve downward. The point OA indicates in both panels the equilibrium $\hat{S}$ that satisfies equation (3). As expected, the open-access equilibrium launches are lower when σ is larger.

1.4. Social Planner

Now consider the case when the orbit is operated by a single agent, which we name the social planner. We assume that the objective of the planner is to choose a launch level S^* so as to maximize the expected net present value of the satellite fleet, $SV(S)$. This requires the first-order condition

$$S^* : V(S^*) = \underbrace{-\frac{\pi}{1+r}\left[q'(S^*)\left(1+\frac{q(g(S^*))}{1+r}\right)+q(S^*)\frac{q'(g(S^*))g'(S^*)}{1+r}\right]}_{\text{External cost of a satellite (EC)}}. \quad (4)$$

Compared to the open-access condition, $V(\hat{S}) = 0$, equation (4) shows that the planner internalizes the effect of an additional satellite (represented here by the infinitesimal change dS) by considering its impact on the collision probabilities faced by the rest of the fleet, which is the external cost term EC in the equation. The external cost has two parts: the first represents the negative impact on the survival probability in the short run (recall $q'(S) < 0$), and the second represents the negative impact on the survival probability between the short run and the long run, captured by the term $q'(g(S))g'(S)$. In particular, the planner correctly anticipates that a collision in the short run generates debris (represented by the factor $g'(S) > 0$), which impacts the probability of survival between the short and the long run, when $\sigma > 0$. The simple model offers an important insight: the external cost of a satellite consists of its impact on survival probabilities from that point onward, taking into account the physical dynamics of objects in orbit, summarized here by the law of motion $g(S)$. Since $V(S)$ is decreasing in S, and the term in brackets in equation (4) is always negative, it follows that $S^* < \hat{S}$. The diagram in figure 1 shows in dashed lines the total social cost of an additional satellite, the PC + EC curve. The curve is always above the PC curve because EC > 0 at all times, and it is downward sloping because the external cost of a satellite declines as the orbit saturates.[11] The planner's solution is denoted by the intersection SP, which corresponds to a lower number of satellites compared to the open-access case. By internalizing the externality imposed by each satellite on the probability of collision, the planner chooses a smaller fleet size than open access.

11. The discountinuity of the PC + EC curve is the consequence the specification we have chosen for $q(X)$, which is not differentiable at $\bar{X}$.

1.5. Kessler Syndrome

A central question in our analysis is how the consideration of economic incentives interacts with the physics of orbiting objects in determining the level of orbit congestion. Specifically, we focus on a particularly undesirable congestion outcome known as Kessler syndrome. Kessler syndrome is defined as a situation in which the current level of objects in orbit implies that the orbit will eventually become unusable in finite time. This level is sometimes referred to as the "runaway growth threshold." In the context of our model, it corresponds to a level of satellite launches that keep the orbit usable in the short run, that is, $S < \bar{X}$, while also making the orbit unusable in the long run, that is, $g(S) \geq \bar{X}$. The idea can be formalized by considering a threshold of objects in orbit in the short run, S_K, such that for any level above the threshold, there is a zero probability of satellite survival in the long run. Using equation (1), the threshold is thus defined by

$$S_K : g(S_K) = \bar{X}. \tag{5}$$

An important question that we consider is whether it is possible for Kessler syndrome to occur in the open-access equilibrium or in the social planner's allocation. Toward answering that question, we first note that in our simple model it must be that $S_K < \bar{X} = g(S_K)$ for Kessler syndrome to be relevant at all. That is, the critical Kessler threshold must be below the level of objects in orbit that make the orbit immediately unusable. If not, Kessler syndrome could never occur as an equilibrium or optimum outcome: since satellites do not earn returns until the short run, rendering the orbit unusable in that period guarantees that the fixed cost of launching will not be recovered. A further implication of this argument is that $S_K < g(S_K)$, which requires a positive net debris generation rate $\sigma > 0$. Intuitively, a runaway growth dynamic can happen only when a collision results in more debris objects than the number of objects involved in the collision, that is, $\sigma > 0$. To see this, suppose that we specify, $q(X) = 1 - X/\bar{X}$, then $g(S) = S(1 + \sigma S/\bar{X})$, and using (5) one can show that $S_K = ((\sqrt{1 + 4\sigma} - 1)/2\sigma)\bar{X}$. Note that $S_K < \bar{X}$ for $\sigma > 0$, and $\lim_{\sigma \to 0} S_K = \bar{X}$, so that Kessler syndrome would only be possible when $\sigma > 0$. A more general version of this condition applies in the more realistic setting of our full model, where debris result from collisions and also decay from orbit over time.

The Kessler syndrome threshold S_K depends on the physics of orbiting objects, summarized here by $q(X)$ and σ. However, whether the threshold is crossed as a consequence of launching choices depends on the economics of orbit use. The following proposition traces the connection between the physics and the economics that are at the center of our analysis.

Proposition 1 (Kessler syndrome): Let the dynamic model of objects in orbit be characterized by equation (1) with $\sigma > 0$, and the Kessler threshold S_K be defined as in equation (5), then

1. under the open-access equilibrium, Kessler syndrome occurs if

$$\frac{\pi}{1+r}q(S_K) \geq F; \tag{6}$$

2. under the social planner allocation, Kessler syndrome occurs if

$$\frac{\pi}{1+r}\left[q(S_K) + S_K q'(S_K)\right] \geq F. \tag{7}$$

The proof is shown in the appendix. Condition (6) states that under open access, equilibrium launches at time 0 exceed S_k and trigger the Kessler syndrome in between the short and the long run when the expected short-run return is higher than the cost of launching. At the individual level, it is optimal to launch because expected profits are positive, even if just accruing in the short run. Condition (7) states that for the social planner, the expected return of the additional satellite launched at 0 has to be corrected for the external cost imposed on the probability of survival of all the existing satellites, which is represented by the negative term $S_K q'(S_K)$. Proposition 1 thus shows that Kessler syndrome is possible under both open access and the social planner but under more stringent conditions for the latter. The diagram in figure 1 shows the Kessler threshold S_K as the dotted vertical line in both panels. In figure 1*A*, the open-access equilibrium is above the threshold, while the social planner solution is not. In figure 1*B*, both open access and social planner are above the threshold. Comparing the two panels shows how the physics and the economics interact: S_K is lower in figure 1*B* because of the impact of a larger σ on the debris growth, but, at the same time, the expected revenue and the expected social cost curves are lower, resulting in both outcomes to be above the threshold.

Using again the functional form $q(X) = 1 - X/\bar{X}$, the condition for Kessler syndrome under open access is

$$F \leq \frac{\pi}{1+r}\frac{1+2\sigma-\sqrt{1+4\sigma}}{2\sigma},$$

while the condition under the social planner is

$$F \leq \frac{\pi}{1+r}\frac{1+\sigma-\sqrt{1+4\sigma}}{\sigma}.$$

Both conditions are more likely to be satisfied if more debris result from a collision (larger σ), the launch cost is lower (smaller F), or if the present value of profits is larger (larger $\pi/(1+r)$).

In summary, the simple model offers two important insights with respect to the Kessler syndrome. First, Kessler syndrome requires autocatalytic debris growth—represented by $\sigma > 0$—that operates in addition to the choice of how many objects to send

in orbit. The reason is that launching intensity endogenously adjusts to the evolution of the collision risk. This is a key distinctive feature of our model compared to physical models with an exogenous launching rate. Second, Kessler syndrome can happen both under open access and the social planner, which means that the full internalization of the external cost of a satellite is not enough to prevent the orbit from becoming unusable. We next develop a fully dynamic model with a more sophisticated physical and economic structure that allows us to expand on the insights offered by the simple model and formulate quantitative predictions.

2. PHYSICS OF ORBIT USE

In this section, we develop a more realistic physical model of orbit use. Consider a spherical shell around the Earth, say the region between 600 and 650 km above mean sea level.[12] There are S_t satellites and D_t debris fragments in the shell in period t, and each satellite collides with another object (satellites or debris) with probability $L(S_t, D_t)$.[13] The laws of motion for satellites and debris are specified as:

$$S_{t+1} = S_t(1 - L(S_t, D_t)) + X_t \tag{8}$$

$$D_{t+1} = D_t(1 - \delta) + G(S_t, D_t) + mX_t. \tag{9}$$

The number of active satellites in orbit in period $t + 1$, S_{t+1}, is equal to the number of launches in the previous period, X_t, plus the number of satellites that survived the previous period, $S_t(1 - L(S_t, D_t))$. Since we assume that active satellites are identical, $L(S_t, D_t)$ represents the proportion of orbiting active satellites that are destroyed in collisions. We assume that the collision probability is twice continuously differentiable, nonnegative, increasing in each argument, and bounded below by 0 and above by 1.[14]

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12. This "shell of interest" approach is frequently used in debris modeling, e.g., Rossi et al. (1998) and Bradley and Wein (2009), though higher fidelity models use large numbers of small regions to track individual objects, e.g., Liou et al. (2004) and Liou and Johnson (2008, 2009b). Paths that span shells are possible and useful for some applications, but highly elliptical orbits (such as Molniya orbits) are the exception rather than the rule. A Molniya orbit has a low perigee over the Southern Hemisphere and a high apogee over the Northern Hemisphere. Molniya orbits require less power to cover regions in the Northern Hemisphere (e.g., former Soviet Union countries) than geosynchronous orbits, due to the low incidence angles of rays from the Northern Hemisphere to geosynchronous positions.

13. It is more convenient to parameterize the physical model in terms of the probability of a collision, L, rather than the probability of no collision, q, as we have done in the simple model in sec. 1. It is of course the case that $q = 1 - L$, and we will use this relationship later in the paper.

14. Satellite operators try to avoid collisions by maneuvering their satellites when possible; the collision probability in this model should be thought of as the probability of collisions that could not be avoided, with easily avoided collisions optimized away. Collisions that could have been avoided but were not due to human error are included in L. Implicitly we are assuming that operators are imperfect cost-minimizers when maneuvering satellites. Even when operators

The amount of debris in orbit in $t + 1$, D_{t+1}, is the amount from the previous period minus the proportion of debris that decay at rate $\delta > 0$, which gives $D_t(1 - \delta)$, plus the number of new fragments created in collisions, $G(S_t, D_t)$, plus the amount of debris in the shell created by new launches, mX_t. Note that the function $G(S_t, D_t)$ plays the role of the debris formation parameter σ in the simple model.[15] We assume that the new-fragment function G is twice continuously differentiable, nonnegative, strictly increasing in each argument, zero when there are no objects in orbit ($G(0, 0) = 0$), and unbounded above ($\lim_{S\to\infty} G(S, D) = \lim_{D\to\infty} G(S, D) = \infty$).[16] We also assume that the effect of the first satellite or debris fragment on new fragment formation is negligible, that is $G_S(0, D) = G_D(S, 0) = 0$ (letting subscripts denote partial derivatives).

For simulations and figures, we use the following functional forms:

$$L(S, D) = 1 - e^{-\alpha_{SS}S - \alpha_{SD}D} \tag{10}$$

$$G(S, D) = \beta_{SS}(1 - e^{-\alpha_{SS}S})S + \beta_{SD}(1 - e^{-\alpha_{SD}D})S + \beta_{DD}(1 - e^{-\alpha_{DD}D})D, \tag{11}$$

where α_{SS}, α_{SD}, α_{DD}, β_{SS}, β_{SD}, β_{DD} are all positive physical parameters. We derive these forms and give some physical intuition about the parameters in appendix B.3 and describe physical model calibration in appendix C.3. Calibrations and scales for figures illustrating qualitative results are chosen to emphasize relevant features.[17] The functional forms do not affect our qualitative results.

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can maneuver their satellites and are aware of impending collisions, coordination can be challenging and plagued by technical glitches (Brodkin 2017).

15. Satellites eventually cease being productive, and their orbits eventually decay without fuel expenditure to maintain their path, features we abstract from in the main model. This abstraction makes some results a little clearer but does not qualitatively affect them. Rouillon (2020) includes these features in the main model and derives qualitatively similar results, albeit in a continuous-time setting. We include limited lifetimes, time-varying payoffs and costs, and state-dependent satellite payoffs in our calibrated simulations in sec. 3.4.

16. While unbounded debris growth is unphysical without an unbounded influx of mass, this is a standard modeling choice in the debris-modeling literature focused on the next few decades and centuries. We provide an argument for this choice over these time scales based on the kinetic energy-object size scaling law and empirical data on fragment sizes following collisions between large intact objects in app. B.4.

17. For figures illustrating qualitative results, S is measured in terms of the number of satellites necessary for the collision probability in a zero-debris environment to be $L(S, 0) = 1 - e^{-\alpha_{SS}S}$. Similarly, D is measured in terms of the number of debris fragments in orbit required for $G(0, D) = \beta_{DD}(1 - e^{-\alpha_{DD}D})D$ new fragments to form. For example, suppose $\alpha_{SS} = 10^{-4}$ and 1 unit of satellites produces a collision probability of 0.01. Then each unit of satellites corresponds to roughly 100 satellites. Similarly, suppose $\beta_{DD} = 0.9$, $\alpha_{DD} = 10^{-5}$, and 1 unit of debris produces 10 new fragments. Then each unit of debris corresponds to roughly 1,000 fragments. The launch rate is measured in units of satellites and inherits its scale from S. The values

3. ECONOMICS OF ORBIT USE

We now present an economic model whose elements match the physical setting described by equations (8) and (9). The key focus is in modeling the economic choice behind the launch rate, X_t. The launch rate will be determined by forward-looking optimizing agents based on the economic institutions (open access or a social planner) and the current state of the environment (i.e., (S_t, D_t)).

We assume that satellites are identical and infinitely lived unless destroyed in a collision. As in the simple model, we assume that the net per-period payoff for a satellite is $\pi > 0$, while it costs $F > 0$ to plan, build, and launch.[18] Operators discount the future at the common rate $r > 0$. Costs and payoffs are assumed to be constant over time and states throughout the analysis. None of our results depend qualitatively on these assumptions. We relax these assumptions in the calibrated simulations in section 3.4, allowing for finite satellite lifetimes, time-varying costs and payoffs, and payoffs that depend on the current stock of active satellites in the shell (e.g., reflecting downward-sloping demand for satellite services).

Finally, we maintain the assumption that a launch rate cannot be negative, that is $X_t \geq 0$. This reflects the fact that there are currently no technologies that can remove a satellite from orbit other than a decision by the operator to deorbit their own satellite.[19] As long as it was optimal to launch a satellite, it will never be optimal for a private operator to deorbit the satellite before its useful life is over. This makes satellites "putty-clay" investments (Johansen 1959): once the satellite is in orbit, it will be operated until it is destroyed in a collision. This generates launch rate policy functions that have an "inaction region" where the launch rate is at its lower bound, that is states (S, D) such that $X = 0$.

3.1. Open Access

3.1.1. Equilibrium

We assume that all economic agents are risk neutral so they are concerned only with expected discounted values. The value at time t of a satellite in orbit, which we denote

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of α_{SS}, α_{DD}, β_{DD} used for these rescalings are obtained from the calibration to actual data described in app. C.3. We scale figures illustrating qualitative results so that $L(1, 0) = (\pi/F) - r$ and so that $G(0, 1) = 0.1$.

18. In reality, satellites accrue one-time (capex) and recurring (opex) costs, while the revenues are recurring annual flows. Satellite capex costs are generally significant. Osoro and Oughton (2021) estimate capex and opex costs for several proposed satellite constellation designs, finding that capex costs are roughly two to four times the size of annual opex costs. For theoretical analysis, it is convenient to write the capex costs separately in F while packaging the recurring revenues and costs into π.

19. Even these decisions may not be actionable, e.g., satellites without propulsion cannot be deorbited on command and must instead wait for natural decay processes to remove them from orbit.

by $Q(S_t, D_t)$, is thus the sum of present payoffs and the expected discounted value of its remaining lifetime payoffs, and it can be represented recursively by:

$$Q(S_t, D_t) = \pi + \frac{1}{1+r}(1 - L(S_t, D_t))Q(S_{t+1}, D_{t+1}). \tag{12}$$

A firm that does not own a satellite in period t decides whether to pay F to plan, build, and launch a satellite that will reach orbit and start generating payoffs in period $t + 1$ or to wait and decide again in period $t + 1$. There is a large number of potential launchers, each one denoted by i. At each period, the launcher i has to decide whether to launch, corresponding to choosing $x_{it} = 1$, or to wait for next period, corresponding to choosing $x_{it} = 0$. If launching, the future value for i will be equal to (12) net of the one time cost F. If not launching, the future value will correspond to the value of a potential launcher at $t + 1$. The problem can be written recursively as

$$V_i(S_t, D_t, X_t) = \max_{x_{it} \in \{0,1\}} \left\{ \frac{(1 - x_{it})}{1+r} V_i(S_{t+1}, D_{t+1}, X_{t+1}) + x_{it} \left[\frac{1}{1+r} Q(S_{t+1}, D_{t+1}) - F \right] \right\}, \tag{13}$$

subject to the laws of motion (8) and (9). The launch rate at time t is then given by $X_t = \int_i x_{it} di$.

In an open-access equilibrium, the launch rate is such that all launching firms earn zero profits, that is, the equilibrium launch rate $\hat{X}_t$ solves

$$V_i(S_t, D_t, \hat{X}_t) = 0. \tag{14}$$

Combining equations (12)–(14), one obtains that in an open-access equilibrium it must be that

$$F = \pi \left(\frac{1}{1 + r - q(S_t, D_t, \hat{X}_t)} \right), \tag{15}$$

where q represents the survival probability of a satellite from t to $t + 1$,

$$q(S_t, D_t, \hat{X}_t) \equiv 1 - L(S_{t+1}(S_t, D_t, \hat{X}_t), D_{t+1}(S_t, D_t, \hat{X}_t)). \tag{16}$$

Equation (15) is the infinite-horizon version of the open-access condition (3) in the simple model. It states that under open access, the equilibrium launch rate equates the cost of launching, F, to the per-period payoff π, discounted by the rate r adjusted by the survival probability of the satellite into next period, $q(S_t, D_t, \hat{X}_t)$. Intuitively, when the survival probability is higher (lower), the effective discount rate $r - q(S_t, D_t, X_t)$ is lower (higher), and thus more (fewer) launches would take place. Note that the survival probability $q(S_t, D_t, \hat{X}_t)$ embeds the physics of orbit use, which impacts the economics of orbit use by adjusting the effective opportunity cost of investing in the satellite asset via (15). At the same time, the choice of investing, represented

by $\hat{X}_t$, impacts the physics of orbit use via the laws of motion (8) and (9), which creates the fundamental feedback between the physics and the economics of orbit use.

3.1.2. Dynamic Analysis of Open-Access Orbit Use

In this section, we explore the dynamics of orbit use under open access, with the goal of highlighting the impact of the economic decision of satellite launchers on the evolution of objects in orbit. We first note that equation (15) can be rewritten as

$$\frac{\pi}{F} - r = L(S_{t+1}, D_{t+1}), \tag{17}$$

which shows that the open-access equilibrium is on an isoquant of the collision probability function—specifically, the isoquant at the level of the "excess rate of return," $\pi/F - r$. Changes in returns, costs, or discounting which lead to the same changes in excess return will have identical effects on open-access behavior. The open-access system is a solution in (X_t, S_{t+1}, D_{t+1}) to equations (8), (9), (17).

Figure 2 shows two examples of open-access trajectories for the launch rate, satellite and debris stocks, and collision probability, under different excess rates of return. In both cases, the launch rate is initially high to ensure that the equilibrium condition holds (i.e., to ensure that the collision probability is on the required isoquant). In subsequent periods, the physical dynamics continue to generate debris, pushing the

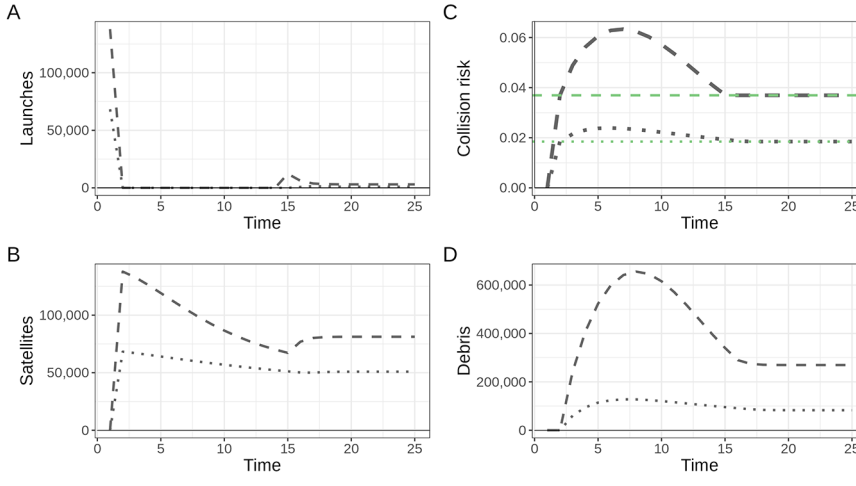


Figure 2. Example open-access trajectories. The dotted line shows a trajectory under a low excess rate of return (*horizontal dotted line* in panel C), while the dashed line shows a trajectory under a high excess rate of return (*horizontal dashed line* in panel C). The initial condition for both trajectories is $(S_0, D_0) = (0, 0)$.

collision probability above the equilibrium level and continually reducing the satellite stock through collisions. The high collision probability forces the launch rate to be zero until the debris and satellite stocks are low enough that a positive launch rate can satisfy the equilibrium condition.

Figure 3 shows an example of the open-access launch policy $\hat{X}(S, D)$ and the corresponding phase diagram. Unlike typical problems in economic dynamics, this problem does not feature a saddle path to the equilibrium or optimum. Instead, there is a basin of attraction to a stable steady state (the "stable basin"). An initial condition will reach the stable steady state if and only if it is within the stable basin. The size and shape of the stable basin is determined both by the system's physics and the economic institution governing launch rates.

Figure 3*A* shows the open-access launch policy. Launch rates are highest when both satellites and debris are low. As either increases, the launch rate decreases. When the satellite and debris stocks are high enough, the launch rate will be zero. Figure 3*B* shows the phase diagram resulting from the physical dynamics of satellites and debris and the launch policy. The curve labeled *Sn* shows the zero-growth isoquant ("nullcline") for satellites and the kinked ellipse labeled *Dn* shows the nullcline for debris. The satellite nullcline takes the shape of the launch policy. As long as (S, D) is to the left of the satellite nullcline, open-access launch behavior will ensure that the stock of satellites is growing.

The "kinked ellipse" shape of the debris nullcline is less obvious. The left side of the kinked ellipse (roughly the portion of the debris nullcline to the left of the satellite

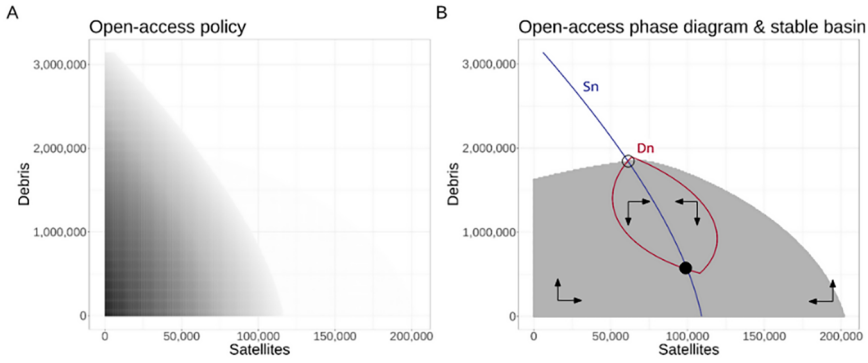


Figure 3. Policy functions, phase diagrams, and stable basin under open access. In panel *A*, darker shading corresponds to higher launch rates. In panel *B*, the kinked ellipse labeled *Dn* shows the nullcline for debris, the curve labeled *Sn* shows the nullcline for satellites, and the gray area shows the stable basin. The arrows show the direction of motion, the solid black circle shows the stable steady state, and the open black circle shows the unstable steady state.

nullcline) results when satellites are being launched. In this case debris growth is the sum of launch debris and what physical processes produce. The right side of the kinked ellipse (roughly the portion of the debris nullcline to the right of the satellite nullcline) results when satellites are no longer being launched. In this case debris growth is solely governed by physical processes. Outside the kinked ellipse there is net positive debris growth, either due to satellite launches or collisions between orbiting objects or both. Inside the kinked ellipse the launch rate and physical dynamics are such that the debris stock will shrink even if the satellite stock increases.

The open-access steady states are pairs (S, D) that satisfy the following conditions:

$$L(S, D) = \frac{\pi}{F} - r, \tag{18}$$

$$X = L(S, D)S \tag{19}$$

$$\delta D = G(S, D) + mX. \tag{20}$$

The intersections of the nullclines in figure 3*B* reveals two steady states satisfying conditions (18)–(20). Proposition 2 shows that this multiplicity is a generic property of open-access orbit use whenever (*a*) active satellites can collide with other active satellites and debris and (*b*) debris objects can collide with each other and produce new debris.

Proposition 2 (Multiplicity and instability): Given a positive excess return on a satellite and a collision probability function that depends on satellites and debris, multiple open-access steady states can exist if debris objects can collide and produce new debris ($G_D > 0$). An open-access steady state will be stable if and only if

$$\underbrace{(\,G_D(S^*, D^*) - \delta)}_{\substack{\text{Net rate of} \\ \text{autocatalytic debris growth}}} \quad < \quad \underbrace{\frac{L_D(S^*, D^*)}{L_S(S^*, D^*)}(G_S(S^*, D^*) + m(\frac{\pi}{F} - r))}_{\substack{\text{Rate of new fragment reduction due to} \\ \text{equilibrium launch activity response to debris}}}. \tag{21}$$

When G is strictly convex in both arguments and two steady states exist, the higher-debris steady state is unstable.

The proof is shown in the appendix. This multiplicity is possible because open access makes the equilibrium satellite stock a decreasing function of the debris stock and autocatalytic debris growth is possible. In a stable steady state, equilibrium reductions in orbit use due to more debris (i.e., the right-hand side of condition [21]) outweigh the incremental debris autocatalysis (i.e., the left-hand side of condition [21]). In an unstable steady state, the reverse holds. When $G_D \equiv 0$, the stable basin fills the entire space. Condition (21) combined with equation (11) shows that when launching decisions are

endogenous the stock of satellites in orbit S^* adjusts and makes it more difficult for (21) to be violated. Even in open access, the private part of the costs of collision risk are internalized, which affects the launching decisions and eventually the number of satellites in orbit. However, once debris have formed there is no endogenous change in behavior that can directly impact the autocatalytic part of their growth, so the economics of orbit use is much less effective in avoiding runaway debris growth. We finally note that proposition 2 is the full model analog to proposition 1, which required $\sigma > 0$ in the simple model for Kessler syndrome (i.e., instability) to emerge.

The number of open-access steady states still depends on the shapes of L and G. For example, if L is strictly increasing and G is strictly convex and smooth, there can be up to two open-access steady states. Only one will be stable. Without smoothness, there may be more steady states with more complex stability properties. Strict convexity of G means that putting more objects in orbit strictly increases the expected number of fragments generated by collisions in each period, for example, due to fragments from one collision propagating and increasing the likelihood of fragment-generating collisions during that period. Smoothness of G means the absence of sharp thresholds in debris formation, for example, due to critical transitions in debris collision dynamics. The shaded area in figure 3*B* shows the basin of attraction to the stable open-access steady state for our example functional forms (eqq. [10] and [11]).

While the possibility of multiple (potentially unstable) steady states in a nonconvex system is a well-understood result in environmental and natural resource economics (e.g., Mäler et al. 2003; Wagener 2003; Lemoine and Traeger 2016), it is less recognized in the growing economic literature on orbit use. While Rao et al. (2020) acknowledge the possibility of an unstable steady state, it is left unmodeled; Adilov et al. (2015) and Rouillon (2020) similarly focus on stable long-run outcomes; and Adilov et al. (2018) do not consider the potential for debris-debris collisions to drive instability. The nonconvexity here is the potential for debris-debris interactions, shown to be theoretically and empirically relevant in the engineering literature on orbit use (e.g., Lewis 2020).

Finally, it is worth asking if the overshooting seen in figure 2—where collision risk first exceeds the equilibrium level before reaching the steady state—is specific to particular parameter sets or a generic property of orbit use models. If open-access equilibria will tend to monotonically approach the stable steady state, then parameter changes that shift the steady state may not cause costly spikes in the debris stock. Such parameter changes include technological advances (e.g., better-shielded satellites), policy guidelines (e.g., encouragement to use frangibolts instead of exploding bolts for booster separation), environmental processes (e.g., sunspot activity that causes δ to vary), or economic changes (e.g., increases in the excess return on a satellite). Proposition 3 shows that the overshooting in figure 2 is a generic property of open-access equilibria: almost all initial conditions will overshoot the stable steady state in satellites or debris or both. Figure 4 illustrates this result.

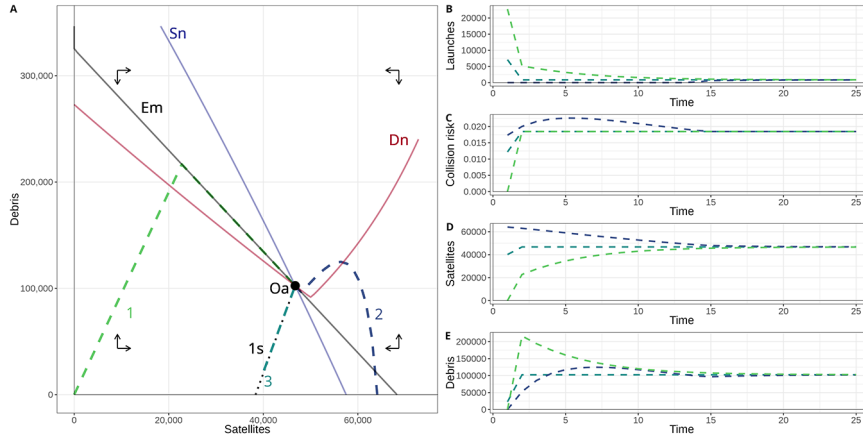


Figure 4. In panel *A*, the thin black line shows the equilibrium isoquant (*curve Em*). The thick black point in panel *A* on the equilibrium isoquant (*point Oa*) is the stable open-access steady state. The dotted black line in panel *A* connecting the steady state to the *y*-axis is the manifold of points that can converge to the stable steady state in one step (the "one-step set," *curve 1s*). Panel *A* illustrates proposition 3 with sample paths. The dashed line labeled "3" along the dotted black line shows a path converging to the steady state in one step. The dashed line labeled "1" shows a path that overshoots in debris and converges to the steady state along the equilibrium isoquant. The dashed line labeled "2" shows a path that initially overshoots in satellites. These paths follow the equilibrium isoquant to the steady state whenever possible, though the physical dynamics force path 2 to approach the steady state from outside the action region. Panels *B*–*E* show the launch rate, collision probability, and satellite and debris stocks for sample paths 1–3 over time. Panel *A* also shows the satellite (*curve Sn*) and debris nullclines (*kinked curve Dn*).

Proposition 3 (Overshooting): Suppose the new fragment formation function is strictly convex in both arguments and the launch rate constraint does not bind. Except on a set of measure zero, open-access paths from initial conditions with positive launch rates will overshoot the stable open-access steady state in at least one state variable.

The proof is shown in the appendix. Figure 4*A* shows three notable features of the open-access equilibrium dynamics. First, curve *Em* is the equilibrium isoquant: a set of points such that the open-access equilibrium condition, equation (17), holds.[20] Paths that eventually converge to the stable steady state (point *Oa*) will attempt to do so along this isoquant. This can be seen most clearly in figure 4C: the lower dashed line representing the path labeled 1 reaches the equilibrium collision risk level even as the satellite

20. The linearity of *Em* is due to the log form assumed for $L(S, D)$.

and debris stocks are away from their steady-state levels. Periods where the collision risk is at the equilibrium level correspond to periods where the path in figure 4*A* is moving along *Em*.[21] Second, curve *1s* is a frontier separating regions of the state space where, along an equilibrium trajectory, (at least) the stock of debris is guaranteed to exceed the steady-state level at some period from regions where (at least) the stock of satellites is guaranteed to exceed the steady-state level at some period. The frontier demarcated by *1s* is exactly the (measure zero) set of points that can converge to the steady state in one step without overshooting in either state variable. Third, there are in general two types of paths to the stable steady state: those that involve positive launch rates every period (exemplified by paths 1 and 3) and those that involve some periods with zero launch rates (exemplified by path 2).[22] The former occur when the total stock of objects in orbit is low enough, while the latter occur when the total stock is large enough.

3.2. Social Planner

3.2.1. Optimum

The value of the satellite fleet and the right to launch new satellites is the sum of present and expected discounted payoffs from satellites in orbit and satellites that are going to be launched. The fleet planner chooses the launch rate to maximize this value:

$$W(S_t, D_t) = \max_{X_t \geq 0} \left\{ \pi S_t - F X_t + \frac{1}{1+r} W(S_{t+1}, D_{t+1}) \right\}, \tag{22}$$

subject to the laws of motion (8) and (9). The planner's optimal interior launch rate X_t^* equates the flow of benefits and costs from a marginal satellite, such that

$$F = \pi \left(\frac{1}{1 + r - q(S_t, D_t, X_t^*) + \xi(S_t, D_t, X_t^*)} \right), \tag{23}$$

where $\xi(S_t, D_t, X_t^*) \geq 0$ is the external cost of a marginal satellite launched in t, corresponding to the right-hand-side term in the simple model condition (4). Equation (23) shows that the planner internalizes the impact of each additional satellite on the survival probability of the satellite fleet by adjusting upward the effective discount rate of per-period returns by the external cost factor ξ, compared to the open-access condition (15). While the general expression for the external factor is difficult to summarize due to the presence of shadow values and differences between future and past periods, the

21. A binding upper bound on the number of satellites launchable per period will cause the path to curve toward the nearest steady state as it approaches *Em*. While this can help firms avoid an unstable steady state following exogenous parameter changes, it does not affect the existence of an unstable steady state.

22. Paths where the launch rate is eventually always zero occur if and only if Kessler syndrome has occurred (see proposition 4 in app. B.5).

steady-state expression for an interior launch rate is more interpretable.[23] Suppressing function arguments and time subscripts, the external cost of a marginal satellite in an interior steady state is

$$\xi = \underbrace{L_S S}_{\substack{\text{“congestion” channel:} \\ \text{cost of satellites colliding} \\ \text{with each other due to crowding}}} \underbrace{+ \frac{1}{1+r}\left(G_S + m(L + SL_S)\right)L_D S + \left(1 - \frac{1}{1+r}\right) m L_D S}_{\substack{\text{“pollution hazard” channel:} \\ \text{cost of satellites colliding} \\ \text{with new fragments and launch debris}}} + \underbrace{\frac{1-\delta+G_D}{1+r}\left(\frac{\pi}{F} - r - (L + L_S S)\right)}_{\substack{\text{“pollution persistence” channel:} \\ \text{cost of persistent debris} \\ \text{and debris growth}}} \tag{24}$$

The first term of ξ, the congestion channel, represents the cost of additional satellite collision probability due to satellite crowding. This is essentially the same externality identified in open-access fisheries in Gordon (1954). As long as the collision probability is coupled with the satellite stock and new satellites weakly increase the probability of satellite-destroying collisions ($L_S \geq 0$), this term is nonnegative. This contemporaneous externality ($L_S S$) could be remedied by coordinating satellites perfectly to avoid each other, for example, using a common slotting architecture for all satellites (Arnas et al. 2021; Lifson et al. 2022).[24] In the general model this corresponds to the collision probability function not having S as an argument, that is, $L(D_t)$ instead of $L(S_t, D_t)$. This form is used in some prior economic literature on orbit use, for example, Adilov et al. (2015, 2018).[25] Since this channel shows only crowding due to active resource users, we refer to it as the “congestion channel” following the distinction between open-access, congestion, and pollution externalities in Haveman (1973). Importantly, note that this channel exists even if debris is irrelevant, that is, $D \equiv 0$ or $L(S, D) = L(S)$ or both—the remaining terms in equation (24) would disappear.[26]

23. We derive eq. (23) and $\xi(S_t, D_t, X_t^*)$ in the general (not necessarily interior, nonstationary) case in app. B.2.

24. Such coordination is technically feasible, but due to the OST it must avoid implying slot ownership.

25. The form in Rouillon (2020) is agnostic on this point, since it focuses on the steady state and treats the collision risk as a “black box” determined by the launch rate. Disambiguating these effects requires more structure on the collision probability function.

26. This cannot be inferred directly from eq. (24) or its more-general form, eq. (96) in the appendix, since both are derived assuming debris exists and affects collision risk. It can be seen instead from the first-order conditions planner’s Lagrangian in eq. (60) of app. B.2. If debris did

Pollution is a different case (Haveman 1973). The residual from production which reduces environmental quality is typically borne by entities other than the residual discharger—typically a firm imposing costs on consumers. Debris constitutes a pollution stock: though the entities bearing the costs may be the same as those discharging debris, its temporal persistence means that future users will be affected by present or past users' choices. Open-access rent dissipation leads firms to ignore these intertemporal effects. Critically, this intertemporal separation means that debris pollution will not constrain equilibrium output via rent dissipation at any given time t except through its effects on the contemporaneous collision risk that operator expects to face. These channels are explicitly noted in equation (24) as the "pollution" channels.

The second term, the pollution hazard channel, is the cost of additional collisions with debris.[27] There are two components of this channel, reflecting debris costs incurred over time and immediately. The first component, $(G_S + m(L + SL_S))L_D S$, represents the cost of colliding with fragments generated by collisions involving other satellites (including fragments from collisions between satellites and launch debris). This component reflects the long-run hazard to the fleet created by the marginal satellite, as it may be destroyed in a collision and generate additional fragments. The second component, $mL_D S$, represents the cost of the new satellite's launch debris—a short-run hazard to the fleet. As the discount rate approaches zero only the long-run component matters, while as it approaches infinity only the short-run component matters. As the number of launch debris fragments from the marginal satellite goes to zero, this channel reduces to the discounted cost of collisions with fragments of satellite-debris collisions ($[1/(1 + r)]G_S L_D S$, a long-run cost) only. As long as having more objects in orbit increases the probability of a collision ($L_D > 0$) and collisions with active satellites or launch debris can produce debris ($G_D > 0$ or $m > 0$), this term is nonnegative.

The third term, the pollution persistence channel, is the cost of debris that does not decay and new fragments produced in collisions between debris objects. If all debris in orbit decayed at the end of each period and could not collide with other debris to create new debris ($\delta = 1$ & $G_D \equiv 0$), this channel would disappear. The cost of this channel is the forgone discounted payoff from a satellite net of collision risk (L), congestion costs ($L_S S$), and the opportunity cost of the funds used to deploy the asset (r). As long as the excess rate of return weakly exceeds the collision probability and marginal rate of congestion costs ($(\pi/F) - r \geq L(S, D) + L_S(S, D)S$), this term is nonnegative.

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not affect collision risk or did not exist, no terms from the debris law of motion would appear in the first-order necessary, complementary slackness, or transversality conditions.

27. By analogy with fisheries, this term reflects damages similar to those caused by "ghost gear"—fishing gear that is lost on the ocean floor and continues to pose a mortality risk to fish (Brown and Macfadyen 2007). We are grateful to one of the referees for suggesting this parallel.

3.2.2. Dynamic Analysis of Optimal Orbit Use

Figure 5 shows two examples of optimal trajectories for the launch rate, satellite and debris stocks, and collision probability, under the same excess rates of return used in figure 2 and from the same initial condition. We induce the change in excess return by lowering the launch cost—the dependence of the external cost of a marginal satellite on returns, private cost, and discounting parameters breaks the symmetry present under open access. In both examples, the launch rate is initially high to take advantage of the lack of debris in orbit. This leads to some overshooting in the satellite stock only (though less than under open access), and the launch rate falls to zero until the satellite stock is closer to its steady-state level. In subsequent periods the satellite stock approaches its steady-state level from above as debris and collision probability grow monotonically. However, the collision probability remains well below the open-access levels. Unlike open access, the optimal collision probability does not overshoot its steady-state level, since the planner internalizes the dynamic externality of launching too many satellites. The planner does, however, overshoot the steady-state level of satellites: given the initial condition $((S, D) = (0, 0))$, the planner takes advantage of "clear skies" before debris accumulates.

Figure 6 shows an example of the optimal launch policy $X^*(S, D)$ and the corresponding phase diagram. As before, there is a basin of attraction to a stable steady state. The optimal launch policy has a similar shape as the open-access launch policy, but a weakly lower magnitude in every state. This expands the stable basin and partially cuts off the "kinked ellipse" of the debris nullcline. Compared to the open-access policy, the optimal policy maintains fewer satellites and less debris in its stable state. The region outside the planner's stable basin in figure 6 corresponds to states where negative launch rates would be required to avert Kessler syndrome; without access to such technologies, the planner is not able to avert Kessler syndrome in those states.

The steady-state satellite and debris stocks are both lower than their open-access levels, the satellite stock by around half and the debris stock by around 80%. As in the open-access case it appears that almost all initial conditions will overshoot the stable steady state in at least satellites or debris or both. Figure 7 illustrates this result.

3.3. Kessler Syndrome

We now define Kessler syndrome in the general setting and show how it can occur under open access or the social planner. Intuitively, Kessler syndrome is a region of the state space from which it is impossible to reach the stable steady state, given the launch policy in use.[28] Definition 1 formalizes this.

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28. In Kessler and Cour-Palais (1978), "collisional cascading" is defined as one of the "possible consequences of continuing unrestrained launch activities." "Unrestrained launch activities" is not defined precisely, though that paper and other engineering studies of the debris environment typically assume either continuation of historical trends indefinitely or continuation with total launch cessation at an arbitrary date (Rossi et al. 1998; Liou 2006; Bradley and Wein

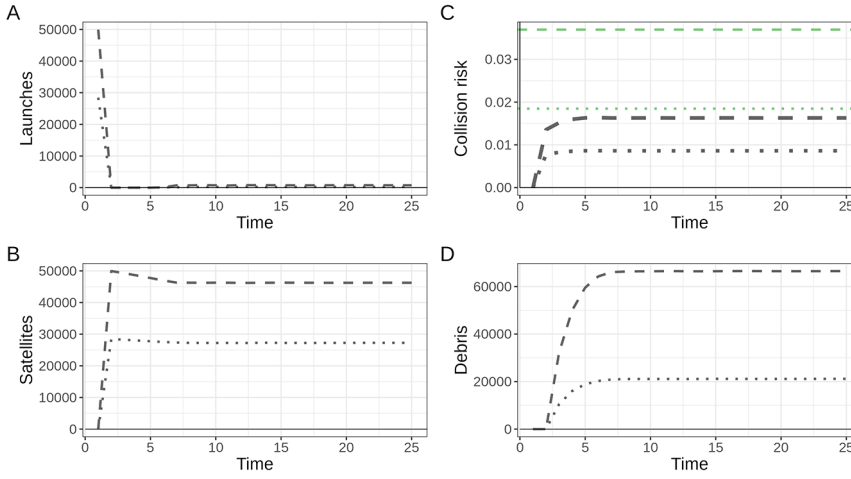


Figure 5. Example optimal trajectories. The dotted line shows a trajectory under a low excess rate of return (*horizontal dotted line* in panel C), while the dashed line shows a trajectory under a high excess rate of return (*horizontal dashed line* in panel C). The initial condition for both trajectories is $(S_0, D_0) = (0, 0)$.

Definition 1 (Kessler syndrome): The Kessler region for a launch policy X is the set of states (satellite and debris levels) such that under X the debris stock will grow to infinity, that is,

$$\kappa_X \equiv \left\{ (S, D) : \lim_{\tau \to \infty} D_{\tau+1} = \infty \middle| S_t = S, D_t = D, X_t = X(S_t, D_t), \tau \geq t \right\}.$$

Kessler syndrome occurs when the system enters the Kessler region, that is, $(S_t, D_t) \in \kappa_X$.

The "Kessler region" in definition 1 is simply the complement of the stable basin. That is, Kessler syndrome occurs when the initial condition places the system outside the stable basin defined by the launch policy and physical dynamics—when (S, D) is outside the gray shaded area in figures 3*B* and 6*B*. Figures 3 and 6 show that the Kessler region is larger under open access than under the social planner, reflecting proposition 1.

2009). Open-access launch behavior is consistent with "unrestrained launch activities," current legal institutions around orbit use, and economic behavior. In Kessler et al. (2010), "runaway debris growth" is defined in terms of a launch policy that holds the stock of "intact objects" (active satellites and unfragmented debris objects) constant. Such a policy is less consistent with rational economic behavior—eventually the required rate of replacement should make launchers prefer to invest their funds elsewhere. In any case, our definition encompasses all such definitions and makes the dependence on the launch policy explicit.

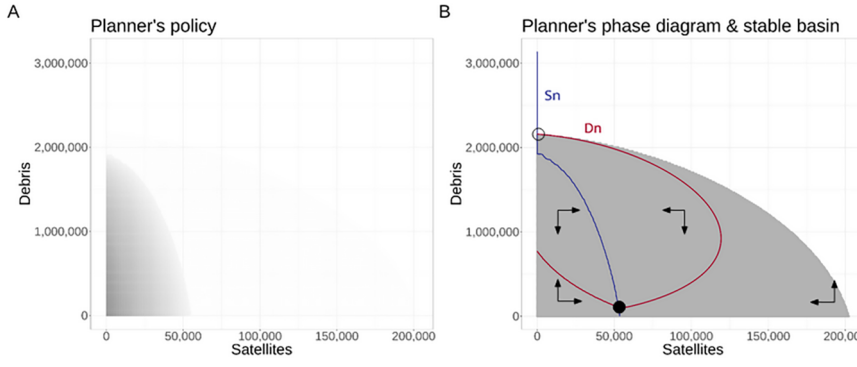


Figure 6. Policy functions, phase diagrams, and stable basin under the planner. In panel *A*, darker shading corresponds to higher launch rates. In panel *B*, the kinked ellipse labeled *Dn* shows the nullcline for debris, the curve labeled *Sn* shows the nullcline for satellites, and the gray area shows the stable basin. The arrows show the direction of motion, the solid black circle shows the stable steady state, and the open black circle shows the unstable steady state.

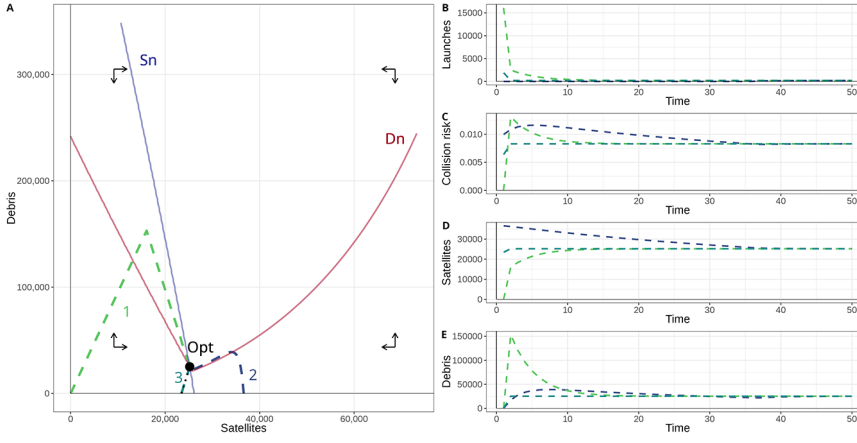


Figure 7. The thick black point in panel *A* labeled *Opt* is the stable optimal steady state. The dotted black line in panel *A* (underneath the dashed line labeled "3") connecting the steady state to the *y*-axis is the manifold of points that can converge to the stable steady state in one step. The dashed line labeled "3" along the dotted black line shows a path converging to the steady state in one step. The dashed line labeled "1" shows a path that overshoots in debris. The dashed line labeled "2" shows a path that starts with more satellites than optimal. Panels *B*–*E* show the launch rate, collision probability, and satellite and debris stocks for sample paths 1–3 over time. Panel *A* also shows the satellite (*curve Sn*) and debris nullclines (*kinked curve Dn*).

The Kessler threshold in the simple model from section 1 now takes the form of a manifold in terms of (S, D) which defines the boundary between the stable basin and the rest of the state space.

The economics of orbit use plays a key role in determining the Kessler region. Given a parameterization of the physical dynamics, the initial condition on its own is insufficient to determine whether Kessler syndrome will occur or not. The economic institutions governing orbit use must be specified to close that loop. Open access and a social planner are two particular institutions consistent with rational forward-looking behavior. The engineering literature tends to assume different institutions, for example, that launches will continue at a rate that maintains the current total population of active objects in orbit regardless of collision risk (Kessler and Cour-Palais 1978) or that the launch pattern from a given interval (e.g., 1997–2004 or 2010–17) will repeat indefinitely (Liou 2006; Lewis 2020). Our model shows that Kessler syndrome projections not grounded in a model of rational forward-looking behavior may miss important behavioral responses. For example, open-access operators will stop launching when the collision risk gets too high, potentially preventing Kessler syndrome. On the other hand, a large-enough increase in satellite profitability may lead open-access operators to rationally choose to cause Kessler syndrome despite prior stability.

Figure 8 illustrates the role of economics in orbital stability through two cases: one where the discount rate is high (fig. 8*A* and fig. 8*C*), and one where the discount rate is low (fig. 8*B* and fig. 8*D*). In all panels, we report two dynamic paths, one starting with a large number of debris and no satellites (dashed arrows) and one starting with zero debris and a large number of satellites (plain arrows). In figure 8*A*, open access does not lead to runaway debris growth for the plain path but does for the dashed path. As the discount rate drops, the open-access stable basin shrinks until it disappears, and Kessler syndrome is guaranteed in all states of the orbit under open access, including the plain one (fig. 8*B*).

For the planner things are different. Under the same parameterization of figure 8*A*, both the dashed and the plain paths are stable and converge to a steady state. As the discount rate increases, the planner places less weight on debris accumulation in the future, which means that the external cost seen in the second and third terms of equation (24) is now lower for all S and D; figure 8*D* shows that this can lead the planner to allow Kessler syndrome along the optimal paths in states where it was not optimal before (dashed line). At the same time, along the stable path (plain line), the planner allows a higher number of satellites and a higher number of debris compared to figure 8*C*.

The planner tends to allow Kessler syndrome (even when they could avoid it by launching less) in states where the short-run return may justify launching, but the long-run returns will be low. Those tend to be states with very few satellites and high numbers of debris. While the planner would not reach those states had they started from a totally empty environment—$(S, D) = (0, 0)$ is within the stable basin—the institutions preceding the planner's control may lead there. The history of LEO use suggests that this may

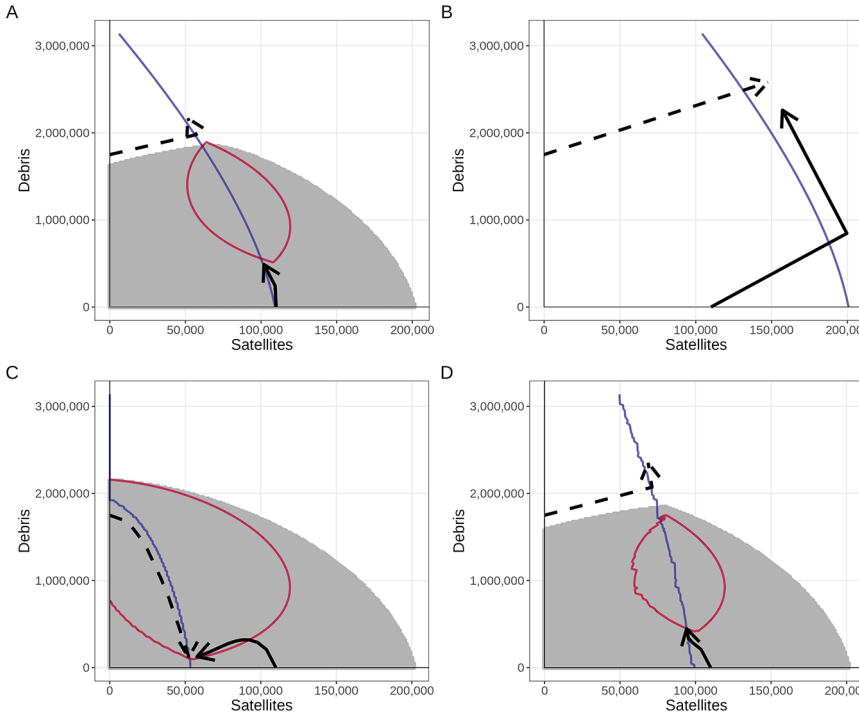


Figure 8. Graphical analysis of emergence of Kessler syndrome for paths for debris and satellites under two different initial conditions: high debris, no satellites (*dashed path*), and high satellites, no debris (*plain path*). The upper row shows the open-access phase diagram and stable basin under two cases of launch costs and discount rates: high cost and low discount rate (low excess return) on the left, low cost and high discount rate on the right. The lower row shows the planner's phase diagram and stable basin under the same parameters. Discretization introduces small irregularities in panel *D*. Kessler syndrome emerges whenever the path is outside the stable basin (*gray area* in the graphs).

be plausible. From the late 1950s to the early 2000s, the majority of LEO users were government entities. This period saw the creation of the bulk of the large debris objects (i.e., rocket bodies and intact payloads) in orbit today, as the United States and USSR (later Russia) conducted activities in LEO primarily for geopolitical and national security purposes. Commercial use of LEO began accelerating in the early 2000s and particularly in the late 2010s, driven by advances in miniaturization, computing, and software. Governments with noncommercial motives have continued to be large debris creators in this period—for example, missile tests by China and Russia in 2007 and 2021—but a growing share of space traffic has been due to commercial operators. Were control of orbital space to be rationalized under a planner's regime tomorrow, the initial condition would reflect decades of geopolitical and open-access use.

3.4. Simulation Exercises

Finally, to get a sense of the plausible range of times till Kessler syndrome occurs (i.e., the system enters the runaway growth region κ_X from definition 1), we calibrate the general model to the 600–650 km orbital shell and simulate open-access behavior forward for 200 years from the 2020 initial condition, $(S_{2020}, D_{2020}) = (158, 626)$. We assume that there are no changes in institutions, no debris removal technologies deployed, and no further anti-satellite missile tests.[29] To account for time-varying payoffs and potential market competition or economies of scale in operating satellites, we rewrite the per-period payoff as $p_t(S_t)$, where

$$p_t(S_t) = \pi e^{at}(1 + \eta)S_t^{\eta}. \tag{25}$$

In the model so far we have taken $a = 0$ and $\eta = 0$, so $p_t(S_t) = \pi$. Equation (25) generalizes this payoff. The payoff consists of two components: factor productivity πe^{at}, and output elasticity $(1 + \eta)S^{\eta}$. The factor productivity parameter πe^{at} represents the productivity of a satellite in an orbital slot at a reference time with exponential growth since then. The elasticity parameter η reflects the occupancy elasticity of satellite output. We also allow time variation in the per-period cost F, which we model as $F_t = ce^{bt}$. We calibrate the parameters p, c such that when $\eta = 0$, $p_t(S_t)$ and F_t match the 2020 aggregate revenues and costs of the satellite sector as reported by the Space Foundation (2021) and the parameter b to match the empirical growth rate of those sectoral costs (2.5%). We also modify the laws of motion to allow for imperfect collision avoidance maneuvers and limited satellite lifetimes. We describe the data and calibration procedures, as well as the modified laws of motion, in more detail in appendix C.

3.4.1. Role of Autocatalytic Growth

The first simulation exercise evaluates the claim that it is the component of autocatalytic growth in the debris growth function $G(S, D)$, represented by the partial derivative $G_D(S, D)$, that primarily matters for Kessler syndrome, as implied by proposition 2. We simulate the model with values of the parameter controlling the new fragments from the debris-debris collision, β_{DD} in equation (11), that range from 0 to 500, and for each, we compute the time at which the system enters the Kessler region, which we refer to as the "Kessler time." Recall that entering the Kessler region does not imply

29. There are several technological and institutional challenges associated with implementing debris removal, including current lack of feasibility and free-rider issues (Weeden 2010; Klima et al. 2016, 2018). Anti-satellite missile tests have been a major source of debris so far (Liou and Johnson 2009a; Oltrogge et al. 2022; Pardini and Anselmo 2023). Although the United States has recently announced a moratorium on their own use of such tests, there is only limited progress toward a binding international agreement preventing their use (Foust 2022; Panda and Silverstein 2022).

that the orbit is unusable but, rather, that the orbit has crossed a threshold beyond which eventual unusability is guaranteed.

Figure 9*A* reports these simulation results. Focusing on the black line, for any value above 450 fragments, the system is already in the Kessler region in 2023, the time of this writing; while for values below 120–150, the Kessler time exceeds 3000. In between, the Kessler time ranges between 2023 and 2130 for β_{DD} between 450 and 400 fragments, with a steep slope that implies a high sensitivity of the stability of the system in that parameter range. For values of β_{DD} below 400, the decrease in Kessler time is more gradual. The middle (dark gray) line in the figure is computed with a value of 332 for the new fragments from satellite-debris collisions, the parameter β_{SD} in equation (11). The two additional curves in the figure are computed with a higher and lower value of the same parameter, 100 and 600. The three curves essentially sit on each other for values of β_{DD} above 200. In particular, they are indistinguishable for values between 400 and 450. The implication is that the Kessler time is very insensitive to the satellite-debris collision part of the debris growth function compared to the autocatalytic part represented by β_{DD}. The numerical results of figure 9*A* thus confirm the theoretical result in proposition 2: in the presence of endogenous launching decisions, Kessler syndrome is primarily driven by the autocatalytic part of debris growth.

3.4.2. Role of Returns Growth and Output Elasticity

The second simulation exercise evaluates the impact of the growth rate of returns (a) and the output elasticity (η) on the emergence of Kessler syndrome. There is as yet insufficient data on the satellite sector to conduct a detailed demand estimation exercise and identify the components of the occupancy elasticity, and the future growth rate of

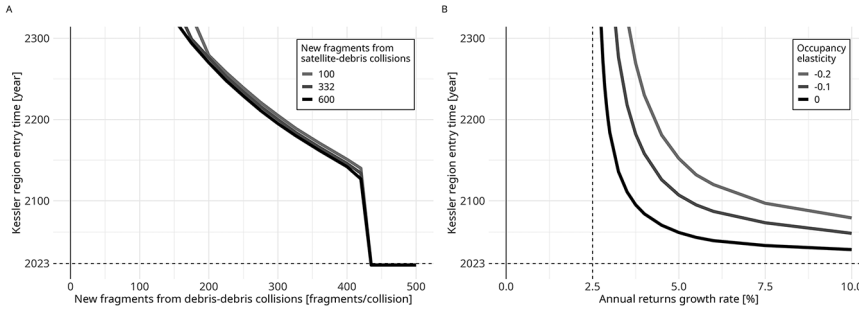


Figure 9. Simulated Kessler syndrome region entry times in the 600–650 km shell under different numerical specifications. Panel *A* plots Kessler times against debris-debris fragmentation, β_{DD} in equation (11), for three values of satellite-debris fragmentation, $\beta_{SD} \in \{100, 332, 600\}$, and with $a = 3\,\%$ and $\eta = 0$. Panel *B* plots Kessler times against unitary satellite returns growth rates for three values of the output elasticity, $\eta \in \{-0.2, -0.1, 0\}$, and with $\beta_{DD} = 326$ and $\beta_{SD} = 332$. In both panels the growth rate of satellite costs is fixed at 2.5%. Details of the calibration are reported in appendix C.

payoffs from satellite operation is uncertain. We therefore conduct a simple sensitivity analysis over a small range of occupancy elasticities, namely, $\eta \in \{-0.2, -0.1, 0\}$, consistent with the range identified in Rao and Letizia (2021),[30] and consider a range of payoff growth rates a between 2.5% and 10%.

Figure 9*B* shows the estimated Kessler times over the annual returns growth rate. Focusing on the black curve, which corresponds to the benchmark case of $\eta = 0$, the results show that Kessler time is highly sensitive to the growth rate of payoffs. For rates above 8%, the Kessler time is between 2040 and 2045, while for rates below 3% it is larger than 2184. The impact of a negative elasticity is represented by the two additional curves, which show that Kessler time is increased across all growth rates, but in a smaller proportion for high growth rates. Specifically, for rates below 3% the Kessler time is 2389 for $\eta = -0.1$, and 2592 for $\eta = -0.2$, while for rates above 8% it is between 2060–73 and 2079–97, respectively. Overall, under the current parameterization, a negative elasticity has a positive impact on the Kessler time because it functions as a dampener on the incentive to launch when the stock of satellites increases. However, we emphasize that this result is not general. As we show in appendix A, a moderately negative elasticity might have a positive effect on the incentive to launch, with an ambiguous impact on the Kessler time. We leave the exploration of the interaction between output elasticity and Kessler time to future work.

4. DISCUSSION AND CONCLUSION

4.1. Discussion

Orbit use bears some similarities to terrestrial resources, though there are also some significant differences. In this section, we briefly discuss some of these similarities and differences. Where possible we refer to the simple model.

4.1.1. The Source of Orbit-Use Externalities

The fundamental coordination problem in orbit use is due to the lack of property rights over orbital slots and the potential for collisions between satellites. The OST thus induces open access to orbit. This open access creates orbit-use externalities due to collision risk, like congestion in a high-seas fishery. Though debris is a source of pollution, its damages are also due to the risk of collisions—not harms to consumers. All orbital

30. Rao and Letizia (2021) estimate a discrete-choice model of sorting over orbits with satellite and debris stocks as orbital characteristics while holding the profile of launches fixed, finding both positive and negative responses to satellites in the same shell for different types of operators—commercial operators seem to prefer to avoid shells with other satellites, while civil government and military operators seem to prefer those shells. However, these substitution and complementarity effects cannot be separately identified in the aggregate η we model without more granular data on revenues and costs for individual LEO satellite operators or a credible identification strategy. To the best of our knowledge the former is not available, and the latter will require further research and is beyond our scope here.

externalities are thus reciprocal and to some degree limited by the same incentives that drive operators to use the resource in the first place. But just as fishers in a high-seas fishery cannot claim exclusive rights over specific fish in a finite population, individual operators in low Earth orbit cannot claim exclusive rights over specific "slots" in a finite (if large) set. Collisional rent dissipation induces resource overuse even as it is limited by expected economic returns (Gordon 1954).

Equation (4) shows the centrality of collision risk in the externality problem: if the collision risk is exogenous, that is, $q(\cdot) \to q$, then $q'(\cdot) \equiv 0$ and the planner's first-order condition matches the equilibrium condition. The externality would disappear as social and private costs align. Slots would be claimed until total revenues equaled total costs and the allocation, whatever risks it entailed, would be economically efficient. But when collision risk is endogenous and marginal satellites do not internalize their effects on the fleet, the open-access equilibrium is inefficient. Note that open-access firms do internalize collision risk directly, but without considering how their actions affect the risks facing others.[31] Again, there is a parallel to open-access fishers accounting for the current stock level due to the effort it implies but not how their harvests change future population growth. While the importance of collision risk can be seen in prior economic literature on orbit use (e.g., Adilov et al. 2015; Rao et al. 2020; Rouillon 2020), the simple model shows clearly how its endogeneity drives the externality even without debris ($\sigma = 0$).

4.1.2. Hotelling's Rule

Hotelling's rule for optimal resource extraction requires that the rate of growth in the profits from resource extraction matches the rate of interest. Here, Hotelling's rule manifests in a requirement that the expected rate of return on capital invested in orbit use equals the discount rate (eq. [17]). There are two main differences between orbit use and resource extraction:

1. Increased orbit use reduces the survival rate of satellites and thus their expected revenues. This effect exists even if it does not decrease the price received by slot users. The problem is also not necessarily finite. In the absence of autocatalytic debris growth, orbital slots cannot be permanently exhausted—filled slots are "replenished" at the net rate of satellite demise and orbital decay.
2. The endogenous risk of collision reduces the rate of return on marginal fleet assets (i.e., the rate of return on the fleet from adding a marginal satellite) below the rate of return on marginal satellite assets (i.e., the rate of return on the marginal satellite added). Thus, while both open-access firms and the planner satisfy a version of Hotelling's rule, the

31. That is, $q(\cdot)$ but not $q'(\cdot)$ is present in eq. (2).

open-access condition (eq. [3] in the simple model, eq. [15] in the full model) considers only a marginal satellite whereas the planner's optimality condition (eq. [4] in the simple model, eq. [23] in the full model) considers the marginal return on the fleet as a whole. A rule similar to the open-access condition here appears in the problem of drilling subject to pressure constraints studied in Anderson et al. (2018), where the decreasing rate of well pressure plays a similar role as the decreasing survival rate here.

4.1.3. Discounting and Renewable Resources

The fact that orbits can be renewed through satellite demise and orbital decay creates similarities to renewable earthbound resources like fisheries. As in earthbound renewable resources, the extent of orbit use under open access exceeds the extent of fleet-wide profit-maximizing use due to the lack of property rights over the underlying natural resource (fish in fisheries, slots in orbits). However, the role of the discount rate differs substantially. Biological assets like fish can reproduce without human investment, allowing a user to profit today at the expense of tomorrow by investing and harvesting many members of the species. Artificial assets like satellites require upfront investment and only generate payoffs over time—investing more today may reduce tomorrow's payoffs due to collisions and will not increase the payoffs received today. Changes in the discount rate thus have the opposite effect on orbit use compared to biological renewable resources. An increase in the discount rate can increase the rate of harvest in a fishery by diminishing the perceived future costs of reduced harvest levels (e.g., see eq. [2.11] in Clark and Munro [1975]) but will decrease the rate of satellite launches by diminishing the perceived future benefits satellites deliver (i.e., increasing r in eqq. [3] or [4] will decrease the number of launches).

4.1.4. Profit-Maximizing Extinction

Finally, Kessler syndrome has similarities to extinction of a biological population. When the discounted future benefits of conservation are below the immediate benefits of exploitation (e.g., the intrinsic growth rate is below the discount rate), profits can be maximized if the species is driven to extinction. As in Clark (1973), even the planner can find it optimal to cause Kessler syndrome if the returns from harvesting the resource over a finite interval exceed the present value of extending the length of the interval indefinitely. However, open-access firms will be "more likely" to cause Kessler syndrome than the planner would. Intuitively, open access means that firms that might receive long-run returns from sustainable orbit use are unable to exclude marginal entrants from launching satellites and crossing the Kessler threshold. Since ceteris paribus larger discount rates reduce the intensity of orbit use, they also reduce the chances of Kessler syndrome—again, a departure from the case of biological populations.

4.2. Conclusion

In this study, we present a dynamic physico-economic model of orbit use under rational expectations with endogenous collision probability and Kessler syndrome. We show how both economics and physics drive equilibrium and optimal orbital-use patterns, derive the external cost of a marginal satellite, and examine conditions under which Kessler syndrome can be an equilibrium or optimal outcome. We highlight three messages regarding orbital-use management.

First, under open access too many firms will launch satellites because they will not internalize the risks they impose on other orbit users. This inefficiency is independent of whether Kessler syndrome is possible or not or even whether debris exists or not. Second, Kessler syndrome is possible when there is autocatalytic debris growth. All else equal, Kessler syndrome is more likely to be an equilibrium outcome as the rate of return on a satellite rises, even if firms respond to orbital congestion by launching fewer satellites. Third, the role of the discount rate in orbit use is unique. Unlike typical bioeconomic commons problems, ceteris paribus higher discount rates induce less (rather than more) open-access and optimal resource use. But if the cost of deploying a satellite is low and the discount rate is high, Kessler syndrome may maximize the net present value of orbit use over the long run, even when all external costs are internalized.

Commons management is one of the oldest problems in economics. Economists tend to favor property rights, corrective taxes, or other market-based mechanisms to solve them. While these mechanisms can ensure efficient orbit use, more data collection and research is needed to understand how orbital-use management policies should be designed in light of its unique features. Whether they are enforced by states, structured as self-enforcing agreements between private actors, or some combination of the two, effective orbital-use management policies must address the open-access problem.

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